CHAPTER 3

Theory and design

This chapter will focus on answering three key questions:

- Why should you choose a mixed methods study?
- What is integration in mixed methods design?
- Which mixed methods design should you choose?

The chapter begins with some general methodological considerations about how we in the social sciences approach our research subjects. Here we further discuss the relationship between theory, evidence and data, which is fundamental to all research, but which becomes particularly central to mixed methods studies. The reason is that in a mixed methods design, it must be very clear how different types of data and analytical methods are connected to the theoretical starting point. This is what we called theoretical integration in chapter 2: that we have a clear theoretical understanding of the phenomenon we are investigating, and that the theory justifies the necessity of collecting different types of data and evidence to shed light on the theoretical ideas about the phenomenon.

It's also the answer to the first question: you should choose to conduct a mixed methods study in order to capture all the aspects of a phenomenon that theory emphasises as important, and not just focus on the parts that one type of data is good at capturing.

A key point in this chapter is therefore that a problem can and should often be operationalised into several research questions to take into account the relevant dimensions of the theoretical basis of the research.

This will often motivate the study to consist of several sub-studies that together can create the necessary evidence to answer the problem formulation. By working with multiple sub-studies and methods, you can gain better and more comprehensive knowledge about the phenomenon than what you can achieve by only looking at one dimension of a phenomenon based on one type of data. The point is that if we take our ideas, theories and concepts seriously, we will often realise that they have multiple dimensions and that a comprehensive coverage requires evidence based on multiple types of data and interpretive methods.

Next, we turn to the chapter's second question about design integration and how to design a study that consists of multiple sub-studies and processes multiple data sets to provide an overall answer to a problem statement. Firstly, we explain what mixed methods design actually means, because when we talk about design integration in mixed methods research, we don't mean the design of the individual study, but rather how the different sub-studies relate to each other and what tasks they each perform in the overall study. We are also talking about the practical organisation, such as whether the sub-studies are conducted simultaneously or in a sequence where one sub-study follows the other.

However, making the purpose of the study clear and explicit is crucial for the optimal study design. We will therefore review the different purposes a mixed methods study can pursue and what a given purpose means for how we should organise our design. In this way, we will also be able to answer the third question, which is which design is right for a given problem formulation.

The motive for mixed methods

Mixed methods is an approach to research that is motivated by a desire to answer a research question in the best possible way, regardless of the types of empirical methods it requires. This may seem obvious, but it's far from obvious in practice. Researchers often select and construct problems based on their methodological speciality: a qualitative researcher trained in interview methods focuses on the part of a given problem that can be solved with qualitative interviews, and makes assumptions about - or completely omits to address - questions about population distributions. It can also be the other way round: the statistical specialist focuses on identifying the correlations that show general relationships in the data, but when it comes to explaining these relationships, the researcher relies on theoretical assumptions of a more or less speculative nature rather than studying the social processes that produce the relationships directly.

In contrast to this approach, the basis of mixed methods research is that *the theoretically motivated problem formulation should guide the research design*, not the researcher's methodological preferences and habits or other considerations. This is the fundamental rationale for integrating multiple methods into one design.

It is important to emphasise that if the problem is best solved by using only one method, then that is the right approach. However, this is not as often the case as one might get the impression from the proliferation of single method studies. A large part of the explanation can probably be attributed to the practice we described above, where there is an initial adaptation of the problem and theory to suit the researcher's methodological preferences.

A mixed methods design therefore starts with the theoretical part of the study, where the problem is defined (Becker, 2020; Bazeley, 2018; Hammersley, 1996; Kelle, 2001). This is also where the first

integration takes place, namely theoretical integration, which consists of establishing a theoretical understanding of the phenomenon that encourages the inclusion of several types of data and evidence to empirically illuminate the problem (Frederiksen, 2013). The theoretical integration thus forms the foundation for us to be able to design a study where several sub-studies can contribute to answering the same problem formulation in a coherent and consistent way.

Theories, evidence and data

The starting point in empirical research is always *theory* or ideas about something in the world. Theories are expressed with concepts and hypotheses that we use to formulate a problem statement: something we do not know or understand about the phenomena we are concerned with. Theories are abstract, and therefore it is only through empirical studies that their validity can be determined. In order to determine if and where a theory is valid, we need *evidence*, i.e. empirical proof that something is a certain way. Evidence is formed through our analysis and interpretation of *data* (Becker, 2020).

It is important to emphasise that data only becomes evidence of something when we have analysed and interpreted it based on our theory about some relationship or phenomenon in the world. Conversely, it is through analysing and interpreting data that we can create evidence for new ideas about the phenomenon and revise or reject the theories we started the study with. Theory, evidence and data are all interrelated and interdependent in the research process. This is true whether we are working with descriptive or explanatory, theory-testing or theory-developing research questions (Alavi et al., 2018).

The unique feature of mixed methods research is that you work with several types of evidence for your theories and often also with several types of data to answer the problem formulation. It is therefore crucial that early in the development of the study's theoretical framework you realise what types of evidence can shed light on the different theories.

concepts and their relationships to each other. It is this realisation that both motivates mixed methods, but also constitutes the fundamental theoretical integration. Often, individual concepts will have multiple dimensions, each of which cannot be captured with the same data.

An example, which we will discuss in more detail below, could be poverty. Poverty can be considered as something objective expressed through statistics or something subjective and experienced expressed through stories and narratives. Single methods cannot provide evidence for both dimensions, so if you want to address poverty in its entirety, theory dictates that you need to combine multiple methods in a mixed methods design. The realisation that the phenomenon of poverty requires evidence from multiple methods is based on the theoretical integration of the objective and subjective sides of poverty.

Research

Types of ~~survey~~ questions

However, it is not enough to have a set of theories to investigate. In order to design the study, you need a set of research questions that operationalise the theories as something that can be empirically illustrated with data. By creating evidence that can answer the research questions, the problem statement and central ideas are illuminated. It is the problem statement and associated research questions that determine how we design our study (Alavi et al., 2018; Becker, 2020). In Table 3.1, we have set out a typology for four types of research questions.

Table 3.1 Typology of research questions

	Theory testers	Theory developing
Descriptive	1. Is X characterised by Y?	2. What characterises X? What is X associated with?
Explanatory	3. Does change in X cause change in Y?	4. What causes Y? How does X cause change in Y?

The rows in Table 3.1 express the first distinction, namely whether the question we ask aims to *describe* or *explain* something. Descriptions aim to gain more knowledge about a phenomenon. This may be because we don't think the existing knowledge is comprehensive or provides a good enough understanding of the phenomenon or takes into account the context in which the phenomenon occurs. Explanation is about understanding how a phenomenon occurs or changes. Here, we are interested in explaining how the phenomenon affects and is affected by other conditions and phenomena.

The columns in Table 3.1 express the second distinction, which concerns the relationship to theory. Do we want to *test* theory or *develop* theory about the phenomenon? Testing can be both descriptions, i.e. what the phenomenon can be said to consist of, and explanations, i.e. theories about how the phenomenon is related to other conditions in the world. When we test, we do so because we have a relatively large amount of knowledge about the phenomenon, which allows us to develop plausible theories and hypotheses about the phenomenon that we find credible. Similarly, we may want to develop theories about what the phenomenon consists of. This is typically something we do when we are unsure if we understand the phenomenon correctly. The development of theory can take place both through descriptions, where we uncover new aspects of a phenomenon, and through explanations, where we develop new theories and hypotheses about its relationships with other conditions in the world.

This typology of research questions is not specific to mixed methods research, but its strength in relation to mixed methods is that it allows you to think about studies that answer and combine multiple research questions based on different types of data and evidence.

Mixed methods in descriptive studies

If we create a descriptive problem statement, we do so with a desire to better understand or map the phenomenon that our ideas address because we don't think the existing knowledge is sufficient or adequate. Let's return to the example of poverty. Poverty can be described statistically by analysing the distribution of income and wealth in the population. Based on a theoretically defined poverty criterion, we can determine whether the number of poor people in the population has changed compared to the previous year. This is a (1) *descriptive, theoretical test* that asks: *Has the number of poor people in the population according to the set criterion increased compared to the number last year*? The static analysis of income data tests for each individual whether they are poor or not, and the data becomes evidence that such and such a number of members of the population are poor in the current and previous year respectively. By comparing the number of poor people in the two years, the testing question of whether there has been an increase in the number of poor people in the population.

Another approach would be to conduct semi-structured interviews with strategically selected people with low incomes and wealth to gain insight into whether they perceive themselves as poor and what their eventual experience of poverty consists of. Here we conduct a (2) *descriptive theory development of* the idea of poverty that answers the question: *How is poverty defined and understood among people with relatively few economic resources, and how can this inform our understanding and definition of*

Poverty? Here, the analysis of the interviews conducted provides evidence for our theory of what poverty is, i.e. we form our idea of poverty based on the evidence. This sheds light on a different dimension of the idea of poverty than the one we shed light on with the statistical analysis above, namely perceived poverty.

Both approaches are valid expressions of different dimensions of poverty, but give us very different insights that do not necessarily match. They can even be directly contradictory, such as if people who are statistically defined as poor do not perceive themselves to be poor. By combining the two approaches in a mixed design, we can thus obtain a description of several aspects of the phenomenon that can complement each other and challenge our theories and provoke new realisations. For example, the statistical analysis' definition of what it means to be poor could be formed based on the interview-based descriptive theory development of how those who have the least in society themselves understand and describe poverty. Conversely, the statistically based descriptive test of who belonged to the group of poor could form the basis for the selection of interviewees who contributed a subjective understanding of poverty through the interviews.

Mixed methods in explanatory studies

Often our problem formulation will be about *explaining* the reasons why a phenomenon looks the way it does or how it relates to other phenomena. Theoretically, the task here is to determine the relationships between the phenomena that our theories are about and determine what evidence best illuminates these relationships and mutual influences. The challenge for explanatory studies is often that methods that collect large-scale data only capture general patterns and relationships at the macro level, not the processes that occur at the micro level.

have created these patterns and relationships. Conversely, small-scale data can capture the actual processes at the micro level, but are challenged in terms of generalising the validity of their findings beyond the empirical field of study and generally seeing the consequences of micro processes at the macro level (Johnson et al., 2019).

Box 3.1 Causal explanations at different scale levels and Coleman's boat

When we talk about explaining the causes of a phenomenon, i.e. causality, it can be useful to distinguish between two dimensions: descriptive causality and productive causality (Johnson et al., 2019). Descriptive causality is typically known from statistics, i.e. large-scale studies where causal conclusions are empirically based on correlations between variables and where we are very certain of the direction of influence. Thus, we are very certain that when one (independent) variable varies in a certain way, it will result in variation in the other (dependent) variable. The limitation of such a large-scale statistical explanation of why variation in one variable affects variation in the other is that it relies on purely theoretical assumptions about the mechanisms and processes that generate the relationship. It is thus the theoretical work of hypothesising that forms the basis for us to go from saying that something is correlated to saying that this is an expression of a causal relationship.

But even with a reasonable theoretical interpretation, we cannot overcome the fact that we are not directly observing the process that produces the patterns expressed by the variables. To directly observe this productive causality, we must use other empirical methods that give us access to detailed studies of actual patterns of behaviour, practices or processes that produce
the causal relationship. To access this type of data, we will typically >>

Use small-scale studies with rich, nuanced and detailed information about a few entities from in-depth interviews, fieldwork or archival studies. The problem with this type of study is that the study of the individual unit is relatively resource-intensive and therefore limited to the small-scale level. With only a few units studied, statistical generalisability becomes difficult to achieve.

A central motive for much mixed methods research is to study both forms of causality, descriptive and productive, each with their own strengths and weaknesses. In this way, it is possible to better provide satisfactory explanations of connections between the micro and macro levels of social change and relationships. This is usually a task that requires combining methods at different levels of scale in a mixed methods study.

This logic has also been articulated by sociologist James Coleman (1990) with a figure that has since become known as Coleman's boat (see Figure 3.1). The causal link between X and Y observed at the structural macro level can only be explained by tracing the process at the micro level of situations and actions. Coleman's point is that what we observe at the macro level is a descriptive relationship that cannot explain itself. Only by studying the micro-level processes that cause a change in X to be associated with a change in Y can we produce a satisfactory explanation.

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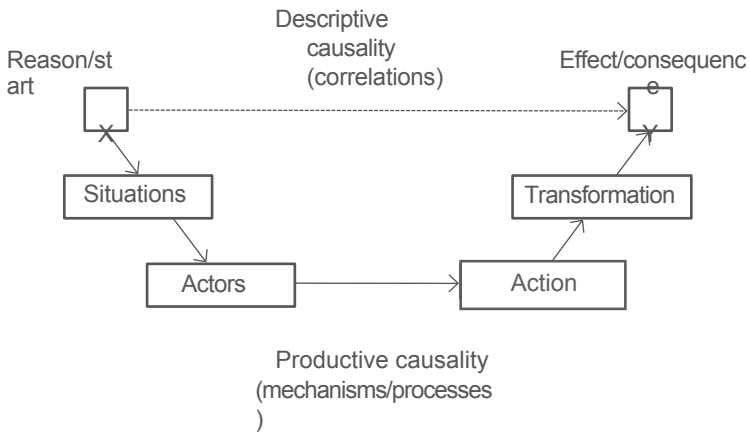

Figure 3.1 Coleman's boat

The obvious answer is to combine small- and large-scale studies of the same context in a mixed methods study. We can illustrate this with a hypothetical example of poverty, which we want to relate to migration patterns. We want to explain poor people's moving patterns and thus ask the (4) *explanatory and theory-building* question: *What explains poor people's moving patterns?* Through statistical analysis, we find that poor people primarily move to areas where many people with low incomes already live. We develop the explanatory hypothesis that housing is cheaper here and therefore economic necessity determines where the poor move to. To (3) *test the explanation*, we conduct a series of in-depth interviews with poor people who have moved or are considering moving. However, in the interviews, no one talks about economic necessity in relation to moving, so our explanation is not confirmed. Instead, we find a different pattern in the interviews, with many interviewees expressing that friendships, family and social

connections are important and that they moved closer to their best friends and family when, for various reasons, their income decreased and they ended up in poverty.

The interview data, which brings us closer to the actual processes, confirms the validity of the patterns we found in the statistical analysis, but shows that the causal mechanism that created the patterns is different from what we first assumed. Here, the interview study has served as both a
(3) *testing an explanation* and (4) *developing a new explanation*. Continuing our imaginary example, we can imagine the theory-developing results from the interview study being formulated as theoretical hypotheses that are statistically tested to examine whether the explanation observed in the interviews is a purely random local phenomenon or has general validity.

Of course, you can also combine survey questions that cross the descriptive-explanatory distinction, which is probably more common than asking two descriptive or explanatory questions. You can also ask multiple questions of the same type, which is less common, but by no means impossible.

We will return to these questions later in chapter 5 on analytical approaches, where they play a central role in relation to the analytical properties of our methods of analysis. What should be clear by now is the opportunities for uncovering multiple aspects and dimensions of phenomena and theories that the mixed methods approach offers, and the central role theory plays in the development of mixed methods design: It is the theoretical considerations about the problem formulation that integrate a theoretical need for multiple types of evidence to answer the various research questions that arise from the problem formulation.

Typical mixed methods designs

But what exactly is a mixed methods study? In line with the definition we introduced in chapter 1, it is a study with several connected empirical sub-studies that solve a problem through integration. The first characteristic is that mixed methods studies consist of several empirical sub-studies that are connected because they address the same problem. The second characteristic is that they are integrated, i.e. they depend on each other in different ways so that the conclusions of the study are based on the results from all sub-studies. The different forms of integration were discussed in chapter 2, and above we have discussed the theoretical integration. In the following, we will deal with design integration: how to organise the sub-studies in relation to each other so that together they best fulfil the overall purpose or motive of the study.

There are a number of basic design types that cover how mixed methods studies are typically organised in relation to each other. They are summarised in Table 3.2. The typology is based on the work of Jennifer C. Greene and colleagues' (1989; 2007) work on mixed methods methodology. This is not an exhaustive typology, so the principles below should not be seen as limiting; only your creativity (and resources) can limit how you design your study.

Table 3.2 Concurrent and sequential designs' confirmatory and complementary logics

Design	Purpose	Sequentially/ Simultaneously
Complementary		
Elaborate	Seeks to strengthen the interpretation and validity of a study by clarifying and elaborating on results across sub-studies. In a sequential design, study 1 can be used to select which part of the study requires elaboration in study 2.	Sequentially and Simultaneously
Developing	Seeks to use results from one sub-study to improve or enable data collection in another sub-study.	Sequential
Initiating	Seeks to produce discrepancies and par- ties between two sub-studies that enable theory development. In a sequential design, substudy 1 can be used to determine which aspect of your evidence can be pro- ductively problematised by substudy 2.	Sequentially and Simultaneously
Expanding	Seeks to use different sub-studies to investigate different parts of a phenomenon so that the study is more complete. In a sequential design, sub-study 1 can be used to locate which part of the phenomenon requires a different method to elucidate.	Sequentially and Simultaneously
Affirmative	The sub-studies' independent measuring points of the same phenomenon confirm or refute each other through a test logic.	At the same time

Notes. In the mixed methods literature, there are a myriad of typologies of purpose and design forms (see e.g. Creswell & Clark 2018; Small 2011; Tashakkori & Teddlie 2010). Our typology is not entirely similar to any of them, but draws on this literature and in particular Greene and colleagues' typology (Greene et al., 1989), which aligns better with our theoretically motivated approach to mixed methods than several of the other typologies, which take a more empiricist approach. The sequential designs are only described with reference to two sub-studies to simplify the logic and maintain clarity. However, there is nothing to prevent the design from consisting of even more sub-studies in longer sequences. Furthermore, it is deliberately not stated whether the sub-studies use one or the other type of data.

Verification and complementarity

Complementarity and confirmation are two fundamental motives for conducting mixed methods studies (Small, 2011; Creswell & Clarck, 2018). When the motive is complementary, the sub-studies are designed to illuminate different elements of the problem. For example, it could be different dimensions of a concept to gain an understanding of multiple aspects of the phenomenon or different parts of a process to gain a more comprehensive understanding. For example, the poverty examples above have all expressed a complementary logic. When doing a complementary design, the goal is to be able to argue claims about multiple aspects of a phenomenon in the social world based on one or more data sources, such as multiple perspectives on poverty.

If, on the other hand, the motive is confirmation, the sub-studies are designed to measure the same thing in order for the studies' findings to confirm or refute each other (Campbell & Fiske, 1959; Webb, Campbell & Schwartz, 1966; Small, 2011). For example, this could be by finding agreement between ethnographic studies of a given context and statistical analyses of the same. This is a test logic. Unlike complementary studies, which aim to produce several complementary claims about the phenomenon, the goal of confirmatory studies is to be able to argue for one claim about the phenomenon with stronger validity because the evidence is based on several different data sources.

Depending on the subject, different requirements are placed on the data design. If the goal is for two studies to confirm each other, it is crucial that the data sources are independent of each other. Data must not be from the same source. This precludes, for example, seeking out some of the respondents from a survey and conducting qualitative interviews with them and then finding that the evidence from both the survey data and the interviews confirm the same idea. The reason why such an approach is unacceptable from a strict confirmatory logic is that it cannot be excluded that the respondents' answers in the subsequent interviews are shaped by their answers to the previous interview.

survey. The two data collection methods thus influence each other, and it is our survey design that fabricates an apparent confirmation. The solution here would be to interview other people who were identical to the participants in the survey in all respects essential to the problem formulation.

Besides the common sources of error we can think of, the idea is that independent data sources should ensure that there are no unobserved common sources of error originating from the same data source. For example, it could be a sampling bias, which means that we got hold of some people from some particular segments of the population that leads us to erroneously confirm a hypothesis with great conviction, as both our sub-studies come to the same conclusion. However, since our problem statement was about all segments of the population, our conclusion is invalid because it only applies to the particular segments that happened to be in the sample. We just don't realise this. If we had used independent data sources, we probably would have found more ambiguous results that could have spurred us to further investigation before jumping to conclusions.

To make a valid confirmation of our findings from the survey, we will need to recruit new participants for our interviews and thus form a new analysis sample that has not been influenced by the survey data collection and is independent of the survey sample. This way, the content of the survey results is tested with less risk of the result being a reflection of either survey influence or sampling bias. In other words, when we seek to confirm our theories about the world, the sub-studies must each be measures of a phenomenon that exists independently of our data. This confirmatory logic is also called *triangulation* (Campbell & Fiske, 1959). It is not used as often today, but it has played a central role in the development of mixed methods research and is still finding new forms, as we discuss in Box 3.2.

Box 3.2 Triangulation - a worn-out metaphor

One of the most important starting points for the mixed methods tradition is the older triangulation tradition (Campbell & Fiske, 1959; Denzin, 1970; Webb et al., 1966). Triangulation is a metaphor taken from navigation, astronomy and surveying, where it is a technique used to determine, for example, the position of a ship at sea (Erzberger & Kelle, 2003). It involves measuring the angle between two known points on land, such as two church steeples. Knowing the angle between the two church towers from the ship's point and the distance between the church towers from your map, the ship's position can be determined with great precision based on Pythagoras' theorem of measuring a triangle where the towers are the two corners and the ship is the last corner.

In social research, triangulation traditionally refers to the idea that multiple points of measurement, i.e. empirical studies of the same phenomenon, can be used to determine the "true" value or meaning of the phenomenon (Campbell & Fiske, 1959). It is thus a position that is based on a realist and positivist philosophy of science. Triangulation exists both in a version that falls within the purely quantitative and purely qualitative paradigms and in a version that cuts across the two paradigms (Flick, 2018; Greene & Caracelli, 1997). The latter version is the one typically referred to today.

Classical positivist triangulation has been heavily criticised and is rarely used today (Hammersley, 2008). This is mainly because it is very difficult to ensure the independence of data sources so that they are credible tests of each other. Not least because the same researchers must be involved to some degree in all sub-studies, at least in the form of sharing a theoretical understanding and knowledge of overall design and purpose. It is therefore unavoidable that the researchers are co-constituting in relation to the outcome of the analysis, and the results thus risk - albeit unintentional - being biased towards a specific purpose of affirmation.

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A number of constructivist and postmodern positions also use the triangulation metaphor, but in a completely different sense. One version is infinite triangulation, where a true definition of a situation is considered impossible and triangulation instead aims to generate multiple perspectives and understandings of the situation without the idea that a complete description is possible (Cicourel, 1974). Another is paradigmatic triangulation, where the goal is also to create paradigmatic oppositions and illuminate a phenomenon from both a positivist/quantitative and an interpretive/qualitative perspective to open up the research field to a more nuanced understanding of the phenomenon. The point here is that the two paradigms cannot be combined and that the phenomenon is constituted in two different ways. The two perspectives do not provide knowledge about the same thing, but construct the same thing in different ways, and the results cannot be integrated into an overall interpretation that provides a more complete picture of the phenomenon (Flick, 1998, 2009).

Finally, Udo Kelle's idea of trigonometric triangulation (Kelle, 2001) should also be mentioned. Kelle's idea is basically that one cannot correctly interpret the results of statistical investigation without insight from small-scale studies into the processes that underlie the observed correlations, and conversely that one cannot understand the processes observed in small-scale studies without insight into the structural framework of the process, which large-scale studies can better map. Kelle's idea is closer to the original meaning of triangulation from navigation, astronomy and surveying, because he insists that the single target from one or the other sub-analysis cannot be correctly interpreted without the insights from the other. Kelle's thinking here is in many ways parallel to Coleman's (1990) thoughts on the relationship between analysing causality at the micro and macro levels as mutually dependent, which we reviewed in Box 3.1.

However, if the motive *is complementarity*, there is no requirement for independent data sources. It is perfectly legitimate to analyse the same data source or create multiple types of data from the same sample in order to investigate different parts of a phenomenon. If there is a robust statistical pattern in the researcher's survey data, but there is doubt about how to interpret it, it makes sense to interview a strategic selection of respondents to get subjective descriptions of the processes behind the statistical pattern and thus obtain an explanation for it. Such a coincidence between survey respondents and interview participants will in fact, and unlike in the case of confirmation, often be a strength when the goal is complementarity. This is why complementary designs often aim to integrate data collection.

In complementary studies, we are primarily trying to learn more about the phenomenon as it appears in our survey data. Consequently, when the goal is to gain complementary knowledge about a phenomenon, it often becomes a strength that the complementary results from the sub-studies originate from the same data source, i.e. that there is data integration. When we can observe the results from the sub-studies in the same observation units, it becomes more reasonable to assume that the results actually express facts about the same phenomenon. The validity of the conclusions about complementary relationships is thus strengthened. Conversely, if we had different data sources for the sub-studies, there could be doubt as to whether the same phenomenon or the same type of process was actually observed in each dataset, and the validity of our conclusions could be more easily questioned.

Complementary designs are by far the most commonly used in mixed methods research. They cover a wider range of purposes than confirmatory studies, which have a relatively narrow definition. Greene and colleagues (Greene & Caracelli, 1997; Greene, Caracelli & Graham, 1989) list four purposes for complementary studies, which we use in this book as a typology of complementary design purposes (see also Table 3.2):

1. The purpose is *elaboration*[1] , when the sub-studies address overlapping, but also different, aspects of the phenomenon. Elaboration occurs when one sub-study illustrates, clarifies or strengthens the findings of another sub-study by adding complementary perspectives and taking into account the mutual weaknesses of the methods (for examples, see Chapter 5, text boxes 5.3 and 5.4, and Chapter 4, text box 4.2).
2. The purpose is *development* when results from one sub-study are used to develop the other sub-study. A typical and widespread variant is when one study acts as a pilot study for the other (e.g. when insights from ethnographic fieldwork are used to develop a survey questionnaire) or forms a selection framework (e.g. when an analysis of statistical data is used as a starting point for quota selection of participants in in-depth interviews). In development designs, there is a built-in sequence where one sub-study comes before and informs the next. Development designs are ideal when you lack the knowledge to conduct data collection in one sub-study if that knowledge can be generated as part of another sub-study (for an example, see chapter 5, text box 5.1).
3. The purpose is *initiation* when the sub-studies are deliberately designed to challenge each other and seek to generate new perspectives, paradoxes and schisms. This approach is often a motive for a mixed methods approach in a field dominated by either the quantitative or qualitative paradigm. This purpose is often linked to a desire to generate new theoretical perspectives on a phenomenon that explain the contradictions (for an example, see chapter 4, text box 4.1).
4. Purposeful *extension* is when you use different methods to measure different components of what you are analysing. For example.

1 Greene and colleagues (1989) actually call this purpose complementarity, but we choose to rename it elaboration here to avoid confusion.

Observation of a production reorganisation intervention in a company and surveys to measure wellbeing and productivity before and after the intervention (for an example, see chapter 5, text box 5.6).

There are of course several dimensions of complementarity, and the important thing here is not to know the above terms. The important thing is to understand the wide range of possibilities, flexibility and creativity that complementary design invites, and that we are moving around in all possible combinations of the types of research questions.

Concurrent and sequential designs

In addition to whether the motive of the mixed methods study is confirmatory or complementary, it is useful to distinguish whether the sub-studies are conducted simultaneously or sequentially. The key distinction here is whether the sub-studies follow each other in a logical sequence, with the results of earlier studies informing later studies, or they are conducted simultaneously, with the results only being collated after the sub-studies have been completed. Table 3.2 summarises the overall design typology based on the two distinctions in terms of purpose (confirmatory/complementary) and the logical organisation of the studies in relation to each other (concurrent/sequential).

If the motive is confirmation, concurrent designs are almost always used, as conducting the studies simultaneously strengthens the assumption of their independence because the results from each sub-study cannot influence the design and thus the results of the other sub-studies. This reduces the possibility of confirmation or disconfirmation being embedded in the design. The types of research questions that are relevant here are found exclusively in the testing questions column of Table 3.1.

For complementary purposes, both sequential and simultaneous designs can be used. Sequential designs are often used as they allow you to utilise the results of the first sub-study when deciding on the

final choices about what evidence the second sub-study should be designed to produce. For example, a developed understanding of statistical relationships can be utilised in the development of an interview or observation guide, and conversely, an intimate understanding of cultural norms and linguistic codes from fieldwork can provide significant advantages in the development of a questionnaire. In both of these examples, the design is sequentially evolving. But one study can guide the next across deepening, initiating, developing and expanding purposes.

However, there can be many situations where a simultaneous complementary design is more advantageous. If you have a strong theoretically motivated expectation of how different sub-studies can contribute to your purpose, a complementary concurrent design can be more flexible and less planning-intensive, as it does not require one sub-study to be completed before starting the next. However, by its very nature, a complementary concurrent design cannot have an evolving purpose, but a deepening, initiating and expanding purpose can be achieved.

Weight

When planning our design, it's important to consider how we prioritise the different data we use in the design. The weight of the design describes different priorities in relation to the role the different data sets play in the generation of evidence. Not all mixed methods projects are designed so that each type of data and sub-study plays an equal role in the final conclusions of the project.

For example, in sequential projects that are designed with a development purpose, the weight between sub-studies is often very different. Here, the first sub-study is often a pilot study with relatively little weight, while the subsequent study is full-scale. The emphasis is therefore on the second sub-study. Here, the pilot study is often theory-developing and it is the theory-developing second sub-study that is the actual main study and the one that will primarily be used as evidence in

the final reporting of

study. This could, for example, be the case where a smaller study with in-depth interviews served to develop an understanding of the given phenomenon, which was subsequently tested through a survey study where the questionnaire was developed based on the knowledge gained from the interviews. In other words, the emphasis here is on the survey.

Another example could be a simultaneous study design where, for example, an intervention is conducted in the form of a field experiment and data is collected on the parameters the intervention is expected to affect before, during and after in the form of repeated surveys. At the same time, field observations have been made during the intervention itself with the aim of gaining in-depth knowledge about how the experiment works. The emphasis is primarily on the main study of the intervention, trying to assess whether there has been a change in behaviour among the experiment participants as a result of the intervention. Less emphasis is placed on the supplementary fieldwork to provide insight into what actually happened during the intervention, so you not only have measurements of whether it worked or not, but also knowledge that can help you understand why and how it worked.

Finally, there may be sub-studies that aim to illustrate or substantiate a point from a main study. These could be interviews or fieldwork that, in a sequential design, illustrate certain correlations found in an analysis of register data. In other words, you can emphasise a particular type of data collection and then supplement it with other types of data.

When the weighting of the sub-studies is unequal, it typically looks a certain way for the different design types in table 3.2. If the aim is exploratory with unequal weight, the secondary sub-study can be used to elaborate and illustrate results and correlations found in the primary sub-study. If you have an exploratory purpose with unequal weight, you typically use the first study with small weight to develop an instrument that

is unfolded in the second sub-study, which has the main emphasis. If the purpose is extension, the emphasis is typically on the first and primary sub-study, while the secondary sub-study is typically used to provide knowledge about a specific problem that the primary sub-study says nothing about, but which is important to the overall problem. On the other hand, if the purpose is initiating, there may be unequal weight if one part of the study is uncontroversial and thus does not require many resources, while the other is controversial and requires many resources. The majority of resources are therefore spent on the controversial part of the study, which should be used to challenge established theories within the research field.

The discussion of weight illustrates a more pragmatic argument in favour of using mixed methods, which concerns the economics of our studies. Different methods have different strengths and weaknesses. For example, although you could conduct 1000 in-depth interviews with randomly selected people in the population and thus gain insight into micro-processes and make a statement about the poverty of the population in a statistical sense, it is a very expensive way to obtain statistical evidence that could be accessed in a cheaper way with available survey or register data. Conversely, one could also try to define all possible explanatory models and test them statistically, including collecting this sea of variables through costly surveys, while it would be far less costly to conduct a series of exploratory interviews to uncover the most obvious explanations.

By extension, but far from a pragmatic argument, there is a final point about the quality of our studies. By quality we mean both the reliability of the ways we seek to measure and capture the social world and the validity of our conclusions. Different data is associated with different threats to the reliability and validity of the evidence that can be analysed. However, these weaknesses can be compensated for through combinations of different methods in designs that explicitly seek to account for and address these threats. Thus, you can

The quality of the study is significantly increased by using mixed methods. We will return to this last point in the following chapters 4 and 5 on data and analysis.

Summarising

In this chapter, we have argued that mixed methods should be seen as an approach to research that dictates that the theoretically motivated problem should guide the research design, not the researcher's methodological preferences and habits. The motive for choosing a mixed methods approach is that you want empirical data that best illuminates the theories about the phenomenon you are investigating.

Next, the starting point for a successful mixed methods design is a well-developed theoretical integration. This means developing a theoretical framework that enables a systematic consideration of the relationship between the study's central theories and the evidence needed to elucidate the theories and the data best suited to capture the social world you are interested in investigating. The mixed methods approach can provide better answers to both descriptive and explanatory problems by shedding light on several dimensions of the same phenomenon or by explaining based on different perspectives on the relationships that underlie the causal explanation.

Next, we have reviewed a number of useful methodological approaches for developing a design that successfully connects theories, evidence and data in a way that achieves the purpose of your study. We have created a typology of the most common designs (see Table 3.2), distinguishing between whether the sub-studies aim to confirm or complement each other on the one hand, and whether they are conducted simultaneously or sequentially on the other. The right choice of design is determined by the problem and the theoretical integration.

Recommended literature:

For more on the relationship between theory and design and specific guidance on how to align design and theory:

- Alavi, M., Archibald, M., McMaster, R., Lopez, V. & Cleary, M. (2018). "Aligning Theory and Methodology in Mixed Methods Research: Before Design Theoretical Placement." *International Journal of Social Research Methodology* 21(5): 527-40. doi: 10.1080/13645579.2018.1435016.

General and easy-to-read introduction to mixed methods designs:

- Small, M. L. (2011). "How to Conduct a Mixed Methods Study: Recent Trends in a Rapidly Growing Literature." *Annual Review of Sociology* 37(1): 57-86. doi: 10.1146/annurev.soc.012809.102657.

Overview and discussion of the design typology used in this book:

- Greene, J. C., Caracelli V. J. & Graham W. F. (1989). "Toward a Conceptual Framework for Mixed-Method Evaluation Designs." *Educational Evaluation and Policy Analysis* 11(3): 255-74. doi: 10.3102/01623737011003255.